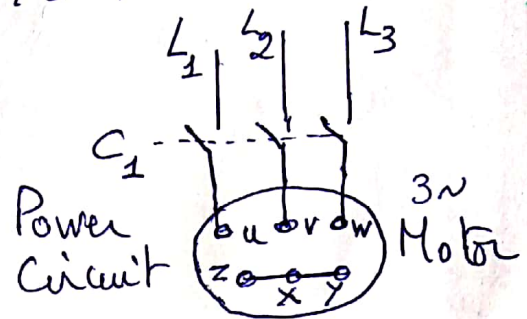
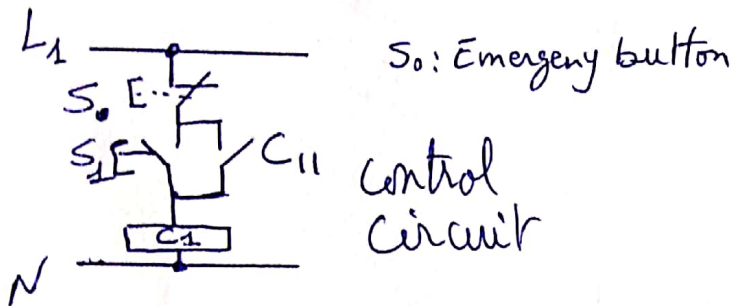


Variation de vitesse - Contrôle (1)

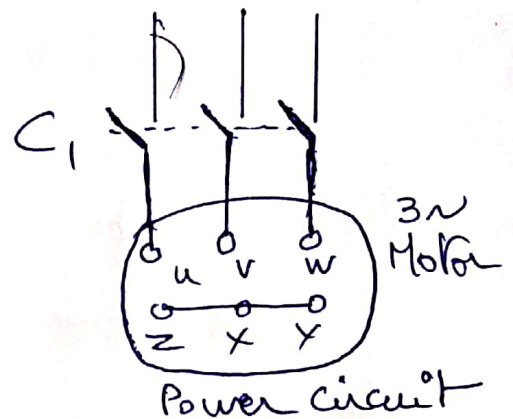
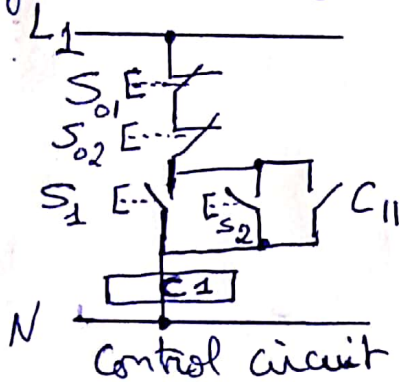
Speed Drive system - Control circuit

Ex. 1: * Commande d'un contacteur : Alimentation d'un moteur électrique via un contacteur

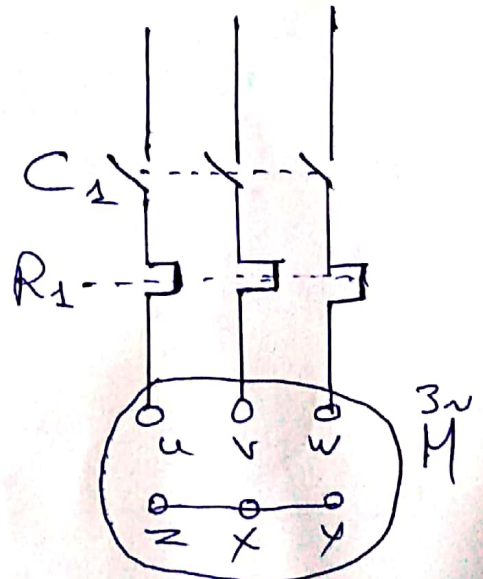
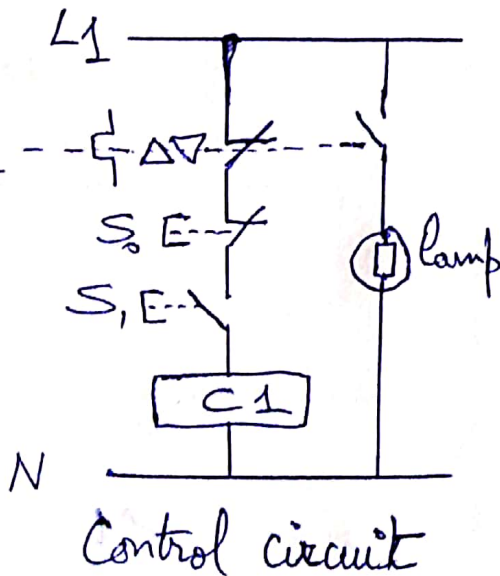
a) * Control of a contactor : Supply of an electric motor via a contactor.



b) Starting a motor by a Contactor at two different places



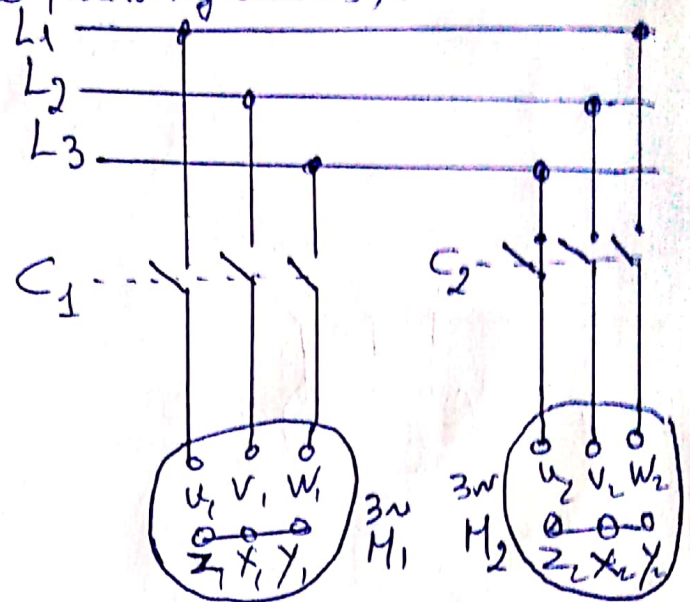
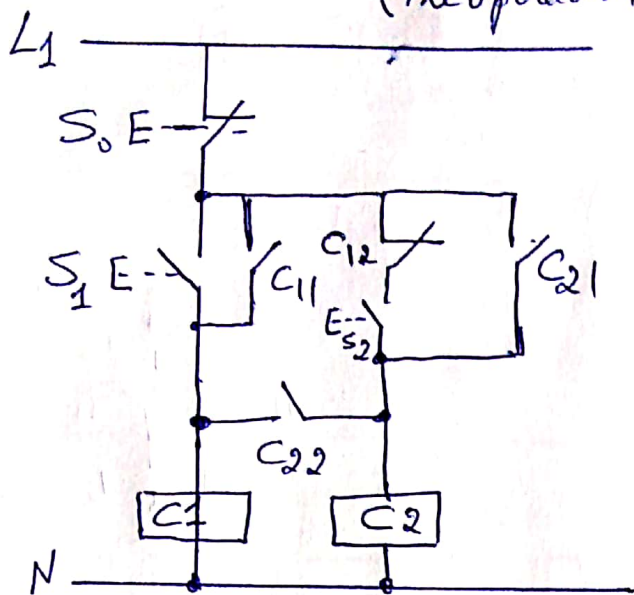
c) Starting a 3~ Motor by a contacteur with thermal relay protection



Speed Drive system - Control

(2)

Ex. 2: Explain how the following circuits work.
(the operation of the following circuits)



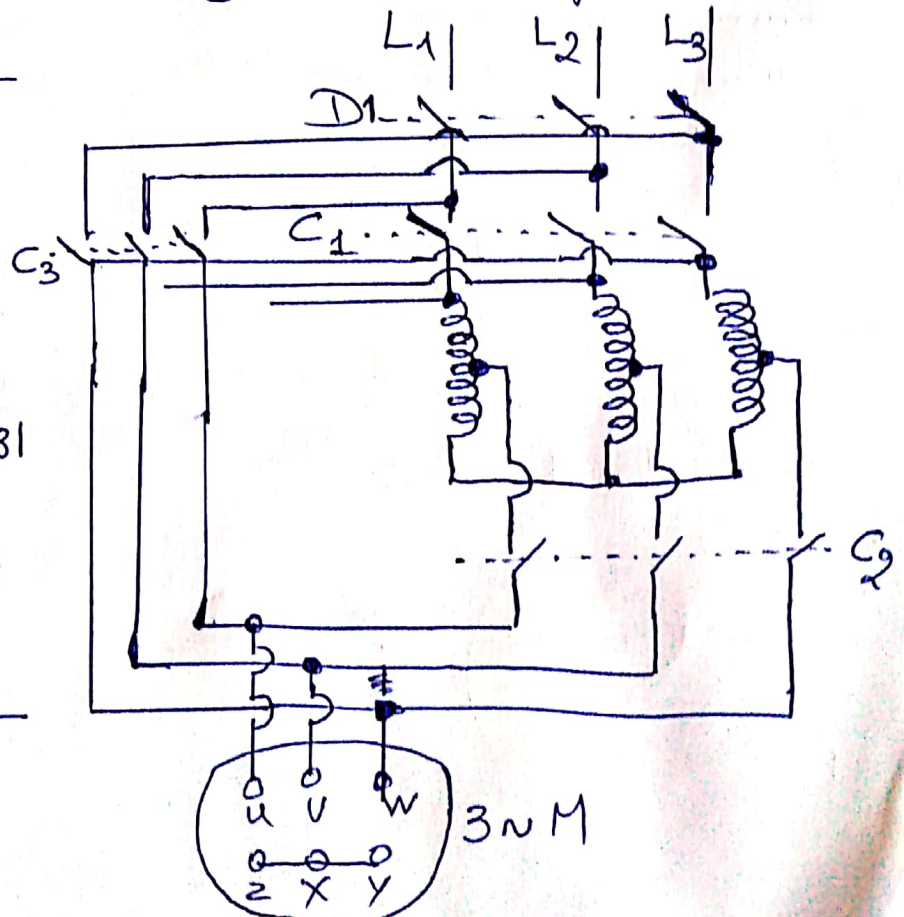
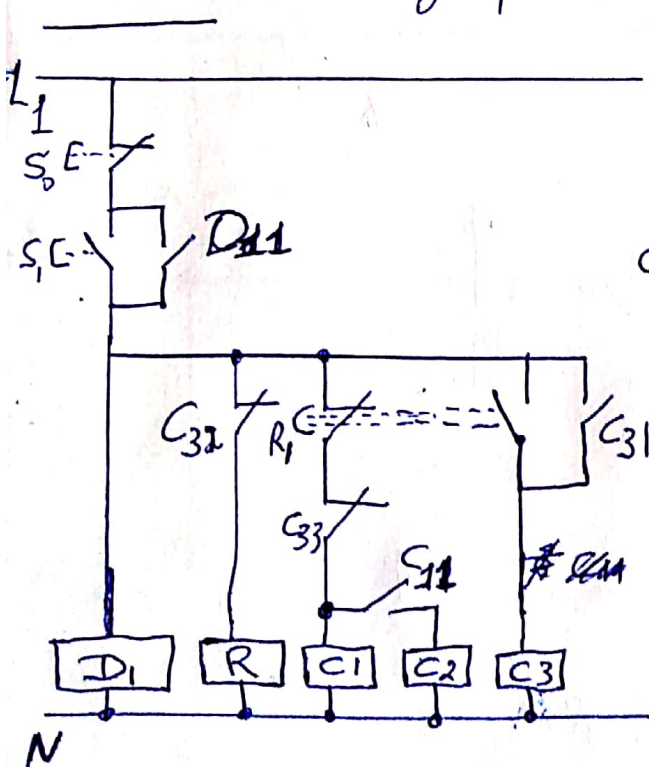
We Press the button

(Push on) $S_1 \Rightarrow$ Coil C_1 ON $\Rightarrow$ Contactor C_1 closed $\rightarrow$ starting of M_1 only

// $S_2 \Rightarrow$ Stop

// $S_2 \Rightarrow$ Coil C_2 ON $\Rightarrow$ Contactor C_2 closed $\rightarrow$ starting of M_1 and M_2

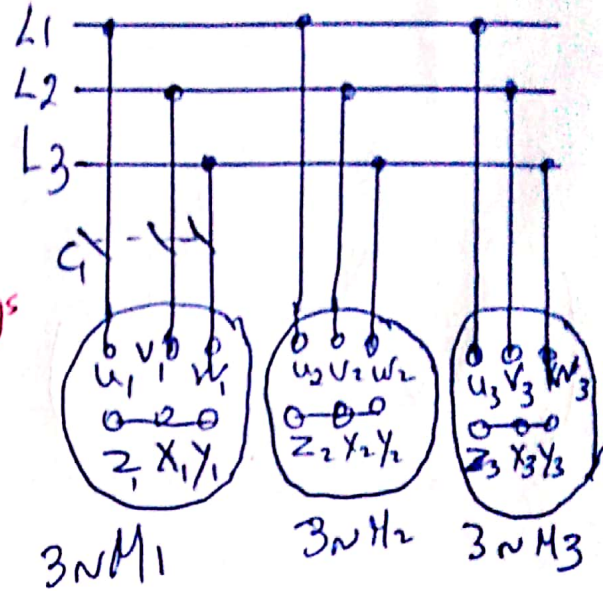
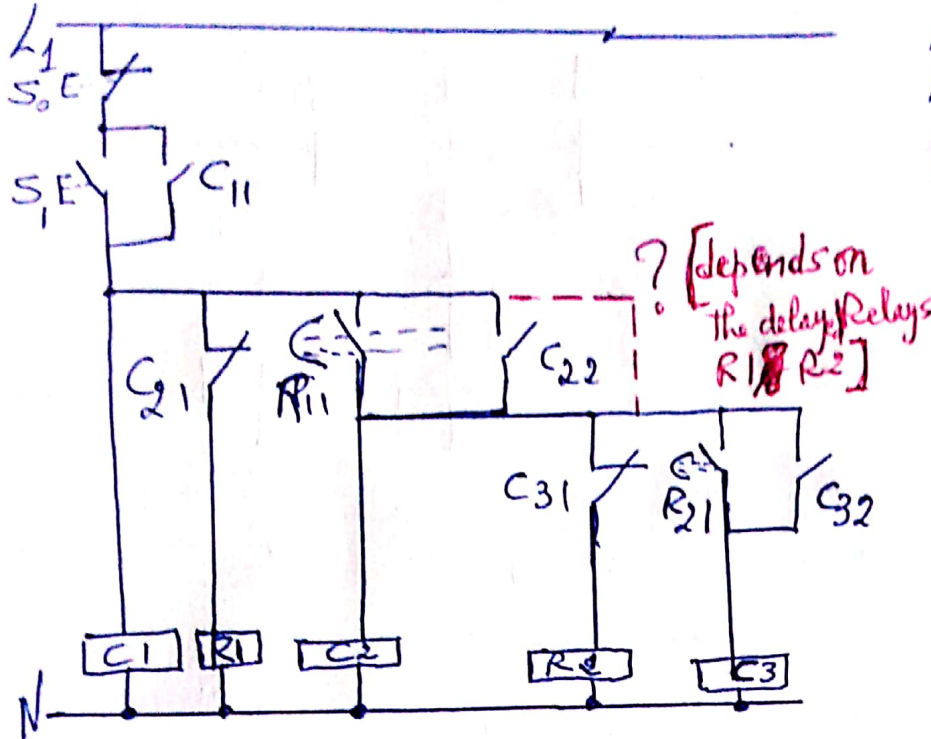
Ex. 3: Starting of 3~ Motor by an autotransformer



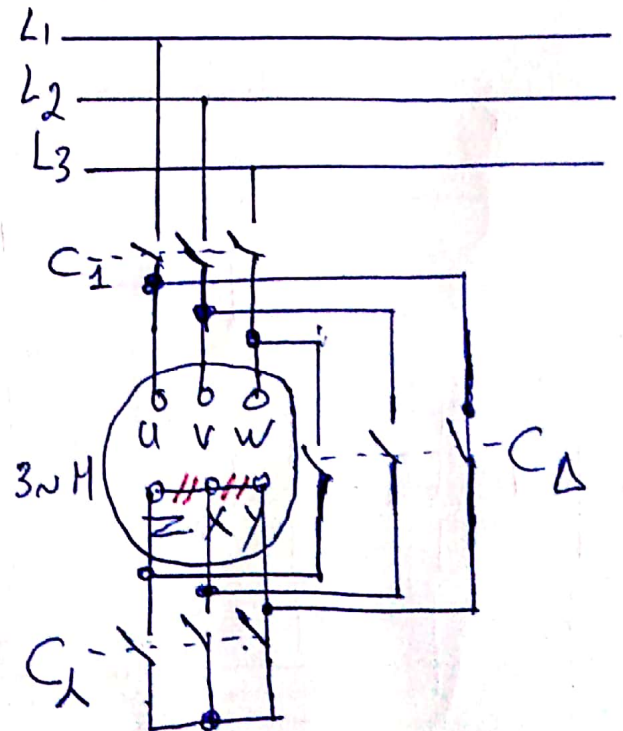
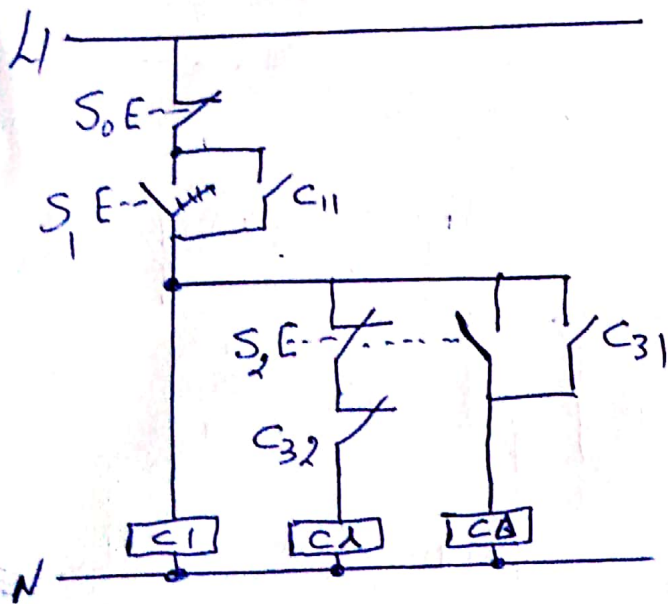
Speed Drive system - Control

(3)

Ex. 4: Automatic starting of 3 motors M_1, M_2, M_3 .
(To avoid the severe transitional regime at start-up)
[otherwise high current peaks]

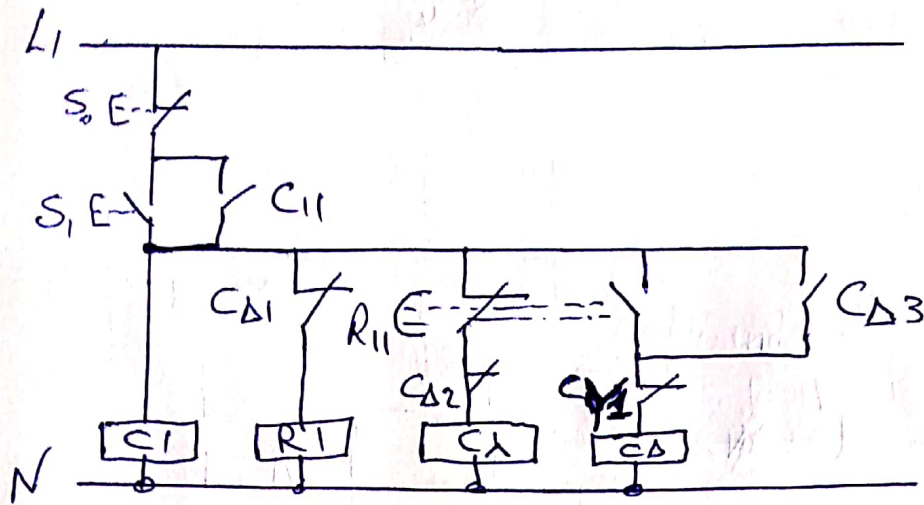


Ex. 5: Manual star-delta starting of a 3 ϕ Motor



Speed Drive system - Control (4)

Ex. 6: Automatic star-delta starting of 3 ϕ Motor



Same Power
Circuit of Ex.5

Ex. 7: Explain the operation of the following circuits.

